CYBERSECURITY PRACTICAL REPORT

FOOTPRINTING, RECONNAISSANCE & NETWORK SCANNING

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Project information	Details
Student / Intern	Jama Abdirisaq
Program / Batch	NetworkWalks Cybersecurity Internship | [B084C]
Reporting period	Week 2 | [11/11/2026]
Modules completed	W2-PM1 – Multiple Kali Linux Tools; W2-PM5 – Zenmap Scanning
Footprinting target	networkwalks.com (assigned practical scope)
Network scanning scope	My own local network / authorized lab network
Authorization	Activities limited to the assigned target and own/authorized network

1. Liability Disclaimer

All activities documented in this report were performed for educational purposes and within the assigned practical scope. Reconnaissance and scanning were limited to the designated target and my own or otherwise authorized network. No unauthorized access, exploitation, or disruption was intended. These techniques must only be used where explicit permission has been granted. The author is responsible for complying with applicable laws, policies, and the agreed scope.

2. Introduction

This report documents the Week 2 practical activities completed as part of the NetworkWalks Cybersecurity Internship. The work covers two modules: W2-PM1, which focuses on footprinting and reconnaissance using multiple Kali Linux tools against networkwalks.com, and W2-PM5, which focuses on network discovery and scanning with Zenmap on my own local lab network.

The purpose of these exercises was to understand how publicly available domain information and network responses can be collected, interpreted, and documented during an authorized security assessment. Screenshots and saved command outputs are included as evidence.

3. Tools Used

Tool	Purpose in the practical
Kali Linux	Environment used to perform the W2-PM1 reconnaissance tasks.
WHOIS	Collect publicly available domain registration and nameserver information.
WhatWeb	Identify publicly observable website technologies and components.
nslookup	Query DNS to resolve the domain and inspect DNS responses.
curl -I	Retrieve HTTP response headers from the target website.
WAFW00F	Check whether a web application firewall is identifiable.
DNSRecon	Collect DNS record information relevant to the domain.
Windows Command Prompt (ipconfig)	Identify the local machine's network configuration.
Zenmap (Nmap GUI)	Discover live hosts on the authorized local subnet and review scan/topology results.

4. Activities Performed

4.1 W2-PM1 – Footprinting & Reconnaissance

The W2-PM1 practical used six tools to collect different categories of publicly observable information about networkwalks.com. The activities and evidence are summarized below.

Task	Tool / activity	Observation to report from your saved output
01	WHOIS	Domain registration / registrar / dates / nameservers
02	WhatWeb	Technologies and versions identified.

03	nslookup	Resolved IP address and DNS response.
04	curl -I	HTTP status and relevant response headers.
05	WAFW00F	Observed WAF result: ModSecurity (SpiderLabs) was identified in the completed run.
06	DNSRecon	DNS record types and relevant values.

The WHOIS task was used to review public domain-registration information. WhatWeb was used to fingerprint website technologies. Nslookup was used to inspect DNS resolution. Curl with the -I option was used to inspect HTTP response headers. WAFW00F was used to identify a detectable WAF, and DNSRecon was used to review DNS records.

4.2 W2-PM5 – Network Scanning with Zenmap

For W2-PM5, the local network configuration was checked using Windows ipconfig. The relevant local subnet was then used in Zenmap to perform the assigned network-discovery scan. The scan results were reviewed to identify responsive hosts and available address information. The Topology view was used to visualize the discovered network where applicable.

5. Risk Analysis / Impact

The following are potential information exposure and network visibility considerations. They are not confirmed vulnerabilities. Final risk ratings should be adjusted to the actual observations and the organization's context.

Finding / observation	Potential impact	Suggested rating
Public domain-registration information is available	May help an observer understand domain ownership and infrastructure relationships.	Low
Website technologies may be identifiable	Technology and version details may help prioritize further security review.	Low–Medium
DNS records are publicly queryable	DNS data may reveal service and mail infrastructure relationships.	Low–Medium
HTTP response headers disclose technical details	May contribute to technology fingerprinting; impact depends on the specific headers.	Low
WAF product is identifiable (ModSecurity / SpiderLabs)	Reveals one aspect of the web application's defensive technology; identification is not evidence of weakness.	Low
Multiple live hosts are visible on the authorized local network	Unexpected devices may warrant inventory and authorization checks.	Context-dependent

These exercises involved information gathering and host discovery. Unless separately tested and documented with authorization, no item in this section should be described as an exploitable vulnerability or a confirmed compromise.

6. Recommendations

1. Review publicly exposed website technology information and remove unnecessary disclosures where practical.

2. Keep the CMS, plugins, operating systems, and network services updated using a documented patch-management process.
3. Review HTTP response headers and configure security-related headers appropriate to the application.
4. Review DNS records periodically and remove obsolete or unnecessary records.
5. Maintain and monitor the WAF; validate its configuration through authorized testing.
6. Maintain an accurate inventory of devices connected to the local network.
7. Investigate unfamiliar hosts through approved asset-management and network administration procedures.
8. Keep network topology and addressing documentation current.
9. Perform reconnaissance and scanning only within an explicitly authorized scope.

7. Conclusion

During Week 2 of the NetworkWalks Cybersecurity Internship, I completed the W2-PM1 footprinting and reconnaissance practical and the W2-PM5 Zenmap network-scanning practical. The W2-PM1 tasks provided practical experience with WHOIS, WhatWeb, nslookup, curl, WAFW00F, and DNSRecon. The W2-PM5 task provided experience checking local network configuration, discovering live hosts with Zenmap, reviewing scan output, and examining network topology.

The exercises reinforced that reconnaissance and scanning are foundational information gathering activities. They also demonstrated the importance of recording exact commands, preserving output, interpreting results carefully, and distinguishing observations from confirmed vulnerabilities. All work must remain within the authorized scope.